

Beyond Aresta's Complex: Ni- and Pd-Catalyzed Organozinc Coupling with CO₂

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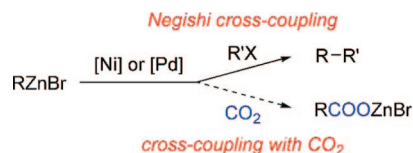
Carbon dioxide is the ideal C1 source for organic synthesis because of its abundance, nontoxicity, and potential as a renewable resource.¹ Thus, designing mild methods to catalytically activate CO₂ and form C–C bonds represents a challenge of both academic and practical importance. Grignard and organolithium reagents are strong nucleophiles which react with CO₂ directly to form valuable carboxylic acid products and derivatives; however, their poor functional group compatibility ultimately limits their use.² Iwasawa and co-workers recently reported a Rh-catalyzed carboxylation of aryl- and alkenylboronic esters.^{2b} Herein, we report a complementary and mechanistically distinct protocol—a Ni- and Pd-catalyzed cross-coupling between organozinc reagents and CO₂.³ Our transformation represents a mild and functional group tolerant strategy for carboxylation.^{2,4} Moreover, this process can be considered an extension of the Negishi cross-coupling to include CO₂ as the electrophilic partner (Scheme 1).

We initially focused our attention on Aresta's complex, Ni(η^2 -CO₂)(PCy₃)₂ (**1**), the first metal–CO₂ complex to be isolated and characterized.^{5a} Aresta's report in 1975 sparked a general interest in using transition metals to activate and fixate CO₂.⁶ Ironically, no method for using **1** in catalytic transformations with CO₂ had been developed.^{5,7} It appeared to us that Aresta's complex (**1**) could undergo transmetalation with organozinc reagents to form organonickel species **2** (Scheme 2).⁸ Reductive elimination from **2** would furnish a new C–C bond, generating a zinc carboxylate and low valent Ni(PCy₃)₂ (**3**), which would bind another molecule of CO₂ through an oxidative cycloaddition to regenerate **1**.

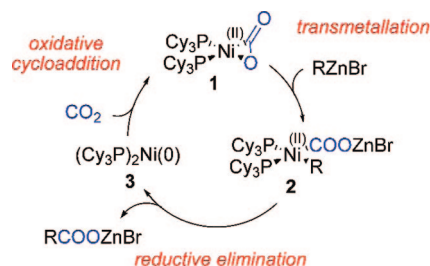
Control experiments confirmed that no background reaction occurs between phenylzinc bromide and CO₂ under mild conditions (Table 1, entry 1). In agreement with our proposal, treatment of PhZnBr with a catalytic amount of **1** (10 mol %) under an atmosphere of CO₂ resulted in the desired benzoic acid in >95% after acidic workup (entry 2). While catalytic fixation of CO₂ by way of insertion to M–C bonds^{2b} or by Lewis acid activation⁹ has been proposed, this is a rare example of catalytic activation by η^2 -coordination of CO₂ to a transition metal.¹⁰ Rather than using the Aresta's complex directly, we examined the use of [Ni(PCy₃)₂](N₂)¹¹ to generate **1** in situ. Under these conditions, the product was formed in >95% (entry 3). For convenience, we also examined the use of commercially available Ni(COD)₂ as a catalyst (entry 4). These results support that complexes **1** and **3** are intermediates in the catalytic cycle.

Due to their high versatility and robustness, we next examined the use of Pd salts as catalyst for CO₂ activation (Table 1). Notably, Pd(η^2 -CO₂)(PMePh₂)₂ has previously been isolated and characterized.¹² We now report that Pd(PCy₃)₂ is catalytically competent,¹² affording carboxylation in high yields (entry 5). Pd₂(dba)₃ as a Pd(0) source was ineffective presumably due to competitive binding of *trans,trans*-dibenzylideneacetone (dba) and CO₂ (entry 6). Indeed, the addition of small amounts of dba completely inhibits the reaction. Pd(OAc)₂ serves as a convenient and easily handled

Scheme 1. Analogy to the Negishi Cross-Coupling



Scheme 2. Proposed Catalysis with Aresta's Complex

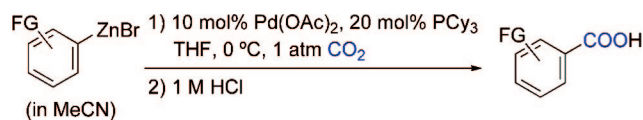
Table 1. CO₂ Activation with Nickel and Palladium

1) x mol% [M], 2x mol% ligand 0 °C, 1 atm CO ₂ PhZnBr (in THF) → 2) 1 M HCl PhCOOH				
entry	x	[M]	ligand	yield (%) ^a
1	0	none	none	0 ^b
2	10	Ni(η^2 -CO ₂)(PCy ₃) ₂ (1)	—	>95 ^c
3	5	[Ni(PCy ₃) ₂](N ₂)	—	>95 ^b
4	10	Ni(COD) ₂	PCy ₃	78 ^b
5	10	Pd(PCy ₃) ₂	—	>95
6	5	Pd ₂ (dba) ₃	PCy ₃	<5
7	1	Pd(OAc) ₂	PCy ₃	>95 (90 ^d)
8	1	Pd(OAc) ₂	P ^t Bu ₂ Me	>95
9	1	Pd(OAc) ₂	PPh ₃	7
10	1	Pd(OAc) ₂	AsPh ₃	2
11	1	Pd(OAc) ₂	TMI ^e	6
12	1	Pd(OAc) ₂	bpy	4

^a Yields determined by ¹H NMR spectroscopy using 1,3,5-trimethoxybenzene as an internal standard. ^b Toluene used as solvent. ^c Toluene-*d*₈ used as solvent. ^d Isolated yield. ^e TMI = 1,3,4,5-tetramethylimidazol-2-ylidene.

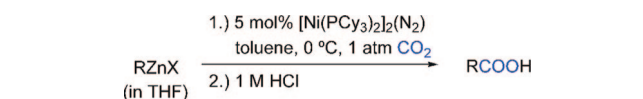
precatalyst (entries 7 and 8), presumably forming Pd(0) in situ.¹³ With this catalyst, use of a simple balloon filled with CO₂ allows efficient carboxylation.¹⁴ A significant ligand effect was observed; PPh₃, AsPh₃, 1,3,4,5-tetramethylimidazol-2-ylidene (TMI), and 2,2'-bipyridine (bpy) ligands resulted in less than 10% of the carboxylated product (entries 9–12). These observations support the idea that electron-rich phosphines (e.g., PCy₃ and P^tBu₂Me) are necessary for oxidative addition to occur between Ni(0) or Pd(0) and CO₂.⁶

Next, we prepared a range of arylzinc bromides by the Gosmini method (Co-catalyzed insertion of zinc into C–Br bonds)¹⁵ to

Table 2. Pd-Catalyzed Arylzinc Carboxylations

entry	functional group (FG)	yield (%) ^a
1	none	90 ^{b,c}
2	<i>p</i> -OMe	90
3	<i>m</i> -OMe	97
4	<i>o</i> -OMe	80
5	<i>p</i> -OAc	97 ^d
6	<i>p</i> -Me	81 ^e
7	<i>p</i> -CF ₃	90
8	<i>p</i> -Cl	88
9	<i>p</i> -F	94
10	<i>p</i> -CN	73
11	<i>p</i> -COMe	75
12	<i>p</i> -COOEt	76 ^e

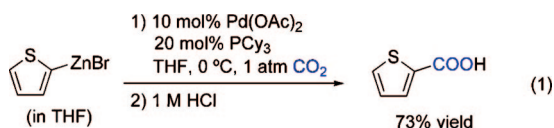
^a Isolated yields. ^b Organozinc solution in THF. ^c With 1 mol % of [Pd], 2 mol % of PCy₃. ^d Isolated as 4-hydroxybenzoic acid after saponification with 1 M NaOH. ^e With 5 mol % of [Pd], 10 mol % of PCy₃.

Table 3. Ni-Catalyzed Aryl- and Alkylzinc Carboxylations

entry	RZnX	yield (%) ^a
1	PhZnBr	74 ^b
2	PhZnPh ^c	80
3	<i>n</i> -C ₅ H ₁₁ ZnBr	90
4	Ph(CH ₂) ₂ ZnBr•LiCl ^d	80
5	Cl(CH ₂) ₂ ZnBr•LiCl ^d	92
6	AcO(CH ₂) ₄ ZnBr•LiCl ^d	75
7	EtOOC(CH ₂) ₄ ZnBr•LiCl ^d	86

^a Isolated yields. ^b With 10 mol % of Ni(COD)₂ and 20 mol % of PCy₃. ^c In situ prepared PhZnEt reacts poorly. ^d See ref 18 for preparation.

investigate the scope of our carboxylation (Table 2). Arylzinc reagents prepared by this route reacted well with Pd(OAc)₂ as the precatalyst,¹⁶ although higher loadings were required. Both electron-rich and electron-poor organozinc reagents were carboxylated in good to excellent yields (entries 1–9), while aromatic rings bearing π -acceptors gave moderate yields (entries 10–12). Notably, substituents at *ortho*, *meta*, and *para* positions can be tolerated (entries 2–4). The carboxylation of arylzinc bromides displayed a significant scope, allowing functionalities traditionally incompatible with Grignard reagents (e.g., ketones, esters, nitriles; entries 10–12).² A heteroaromatic reagent could also be effectively carboxylated (eq 1).



Complimentary to our Pd-based protocol,¹⁷ we found that, under Ni catalysis, both aromatic and aliphatic zinc reagents could be tolerated (Table 3). Commercially available PhZnBr, Ph₂Zn, and *n*-C₅H₁₁ZnBr underwent CO₂ incorporation to produce the corresponding carboxylic acids in good yields (entries 1–3). Alkylzinc reagents (prepared by Knochel's method)¹⁸ bearing various functionality (e.g., Ph, Cl, AcO, EtCOO substituents) were also well-tolerated (entries 4–7). In contrast to Iwasawa's Rh-catalyzed

method,³ this approach enables carboxylation of functional group compatible aliphatic nucleophiles.¹⁹

In summary, we have presented a novel catalytic strategy for carbon dioxide incorporation. Aresta's complex can catalyze the cross-coupling of organozinc reagents with CO₂, and Pd(OAc)₂ is shown to be a convenient catalyst precursor for carbon dioxide activation. We are further investigating the mechanism of this process and plan to use this concept toward asymmetric carboxylation.

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Supporting Information Available: General procedures for carboxylations, organozinc synthesis, catalyst preparation, spectroscopic data. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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